DM – Basic counting

NGUYEN Hoang Thach
nhthach@math.ac.vn

Outline

1. General information
2. Basic principles of counting
   - The product rule
   - The sum rule
   - The subtraction rule
   - The division rule
3. Pigeon hole principle

- **Course description:**
  - 9 weeks $\times$ (2h lecture + 2h tutorial)
  - two parts: *counting* and *introduction to graph and tree*
- **Evaluation:**

$$0.1 \times \text{attendance} + 0.3 \times (\text{assignments and/or tests}) + 0.6 \times \text{exam}$$

- **Textbook:**
  - K. H. Rosen. *Discrete mathematics and its applications, 7th ed.*, McGraw-Hill, 2012.
  - *For the curious:*
    Graham, Knuth and Patashnik. *Concrete Mathematics, 2nd ed.*, Addison-Wesley, 1994.

The importance of counting

The counting problem:
What is the number of objects having certain properties?

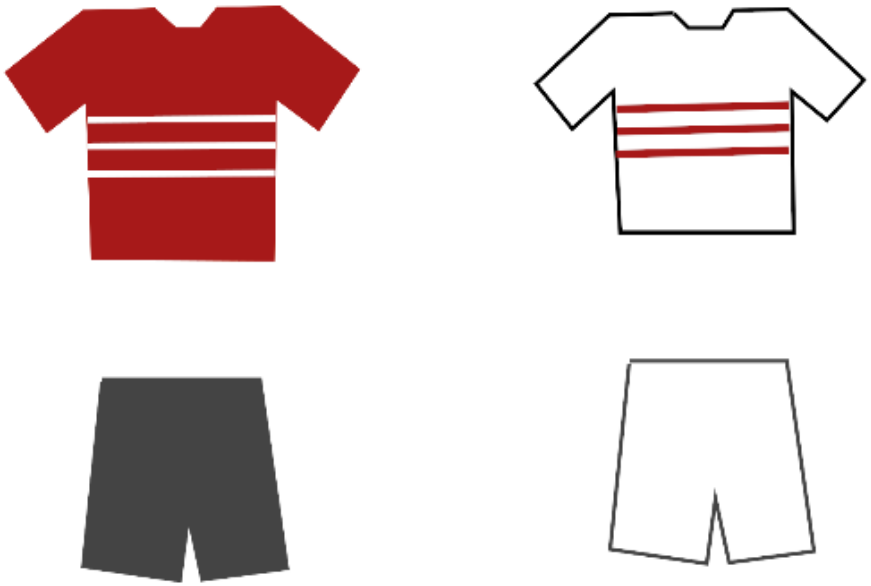
Examples:

- Determining the length of telephone numbers in a country.
- Estimating the (time) complexity of an algorithm.
- Estimating the storage size needed by a database.
- Evaluating probabilities.
- etc.

Outline

1. General information
2. Basic principles of counting
   - The product rule
   - The sum rule
   - The subtraction rule
   - The division rule
3. Pigeon hole principle

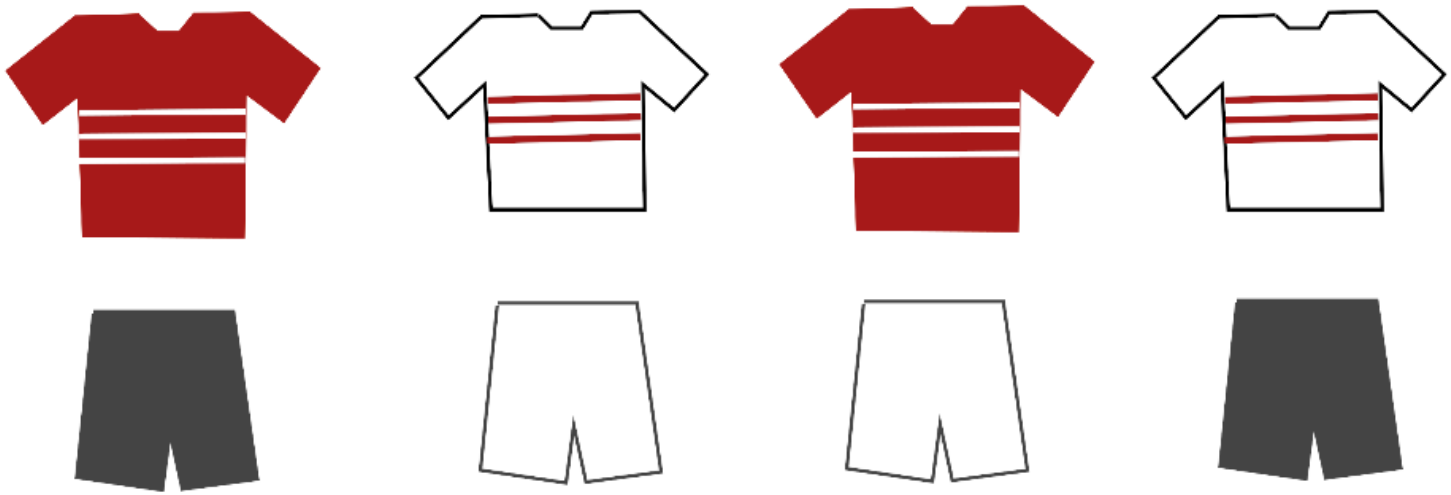
A football club's kits



How many different kit combinations are there?

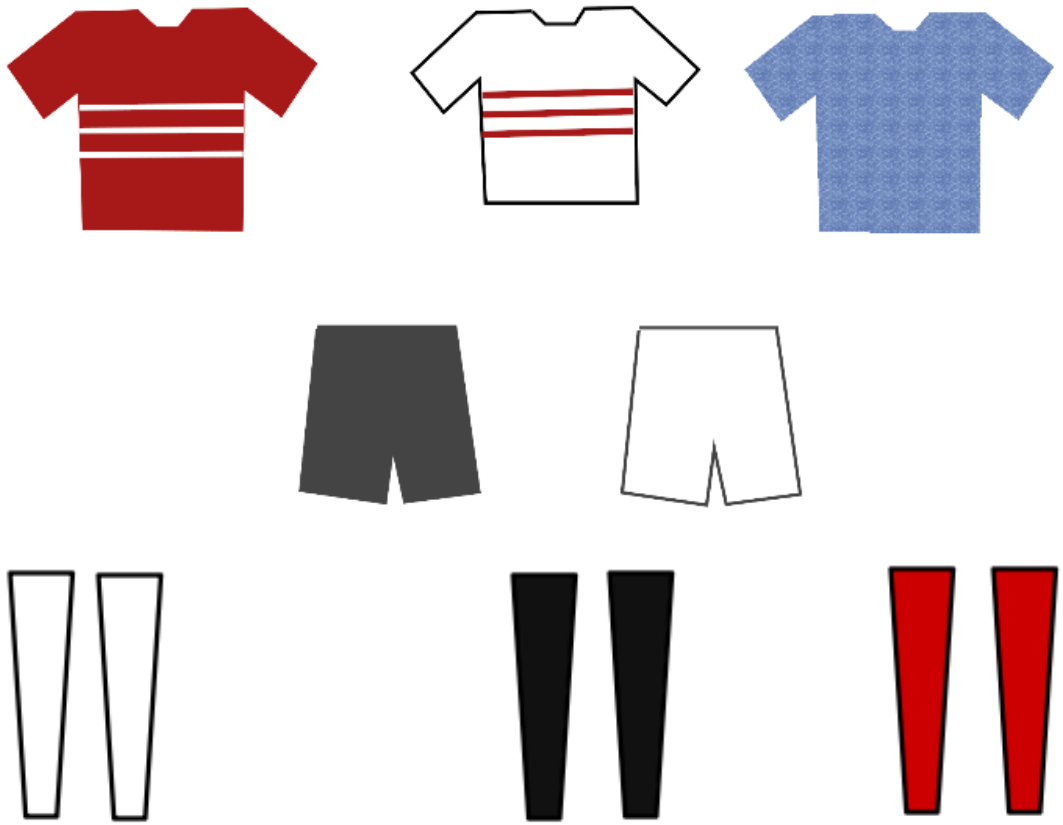
A football club's kits

Naive way: Listing all possibilities.



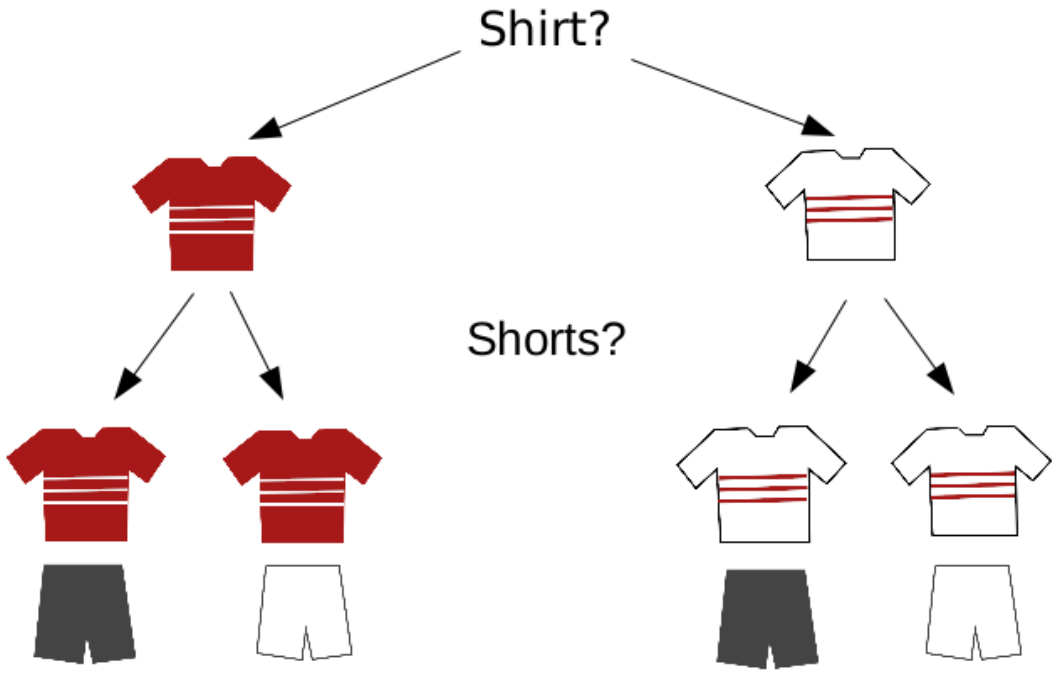
A football club's kits

Naive way: Listing all possibilities?



A football club's kits

Choosing step-by-step



Step 1: Choose shirt. There are 2 choices.

Step 2: Choose shorts. *For each* choice of shirts, there are 2 choices for shorts.

$$2 \times 2 = 4 \text{ choices}$$

The product rule

When and how does the *product rule* apply?

- To count the number of ways to carry out a *sequential* procedure.
- The number of choices at each step of the procedure is the same regardless of the previous choices.
- The total number of ways equals the product of the numbers of ways to do each step.

The product rule – example

Example:

The seats in a theater are labeled with an uppercase letter followed by an integer not exceeding 50. If each label is used exactly once, then how many seats are there in that theater?

- Each seat label has the form "[A-Z][1-50]".
- **Step 1:** Choose the letter. There are 26 ways.
- **Step 2:** Choose the number. There are 50 ways.

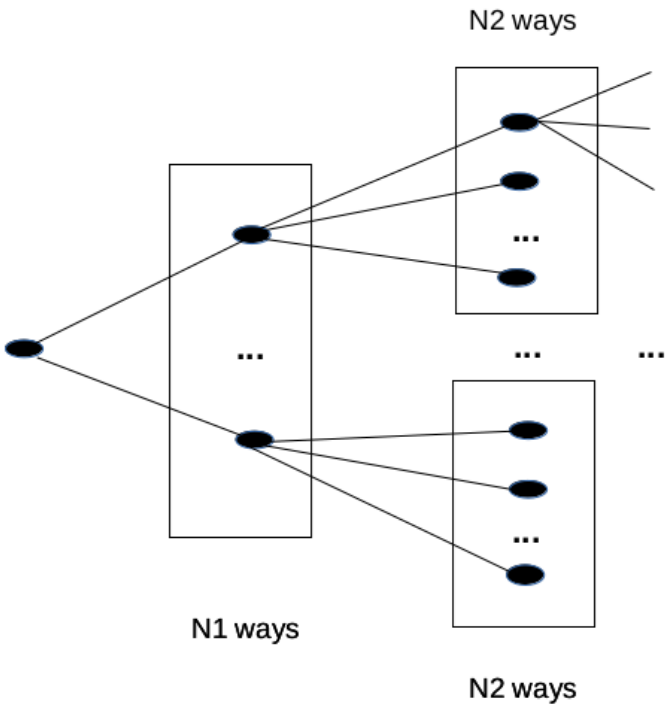
Ans: $26 \times 50 = 1300$.

The product rule – a formal description

Definition (Product rule)

Suppose a procedure can be broken down in to k steps. Suppose that there are n_1 ways to do step 1, n_2 ways to do step 2, . . . , n_k ways to do step k. Then there are $n_1 n_2 \dots n_k$ ways to do the procedure.

$$|A_1 \times \cdots \times A_k| = |A_1| \times \cdots \times |A_k|$$



The product rule – examples

Example:

A variable (in a program) is stored using 7 bits. How many different values can that variable have?

Step i**:** Choose the value for bit i (0 or 1).
By the product rule, there are $2^7 = 128$ different values.
($k = 7$, $n_1 = n_2 = \cdots = n_7 = 2$)

The product rule – examples

Example:

What is the number of subsets of the set $S = \{1, 2, \ldots, n\}$*?*

Each subset can be determined by an n-step procedure: step i decides whether the element i belongs to the subset.

There are two choices for each element, hence by the product rule, there are 2^n subsets of S.

Remark: Another way to count the number of subsets is to identify each subset A with a binary string s of length n: the i-th bit of s is 1 *iff.* $k \in A$. Since there are 2^n binary strings of length n, there are 2^n subsets.

The product rule – examples

Example:

What is the value of k after the execution of the following nested loop?

$k := 0$
for $i_1 := 1$ **to** n_1
  for $i_2 := 1$ **to** n_2
    .
    .
    .
    for $i_m := 1$ **to** n_m
      $k := k + 1$

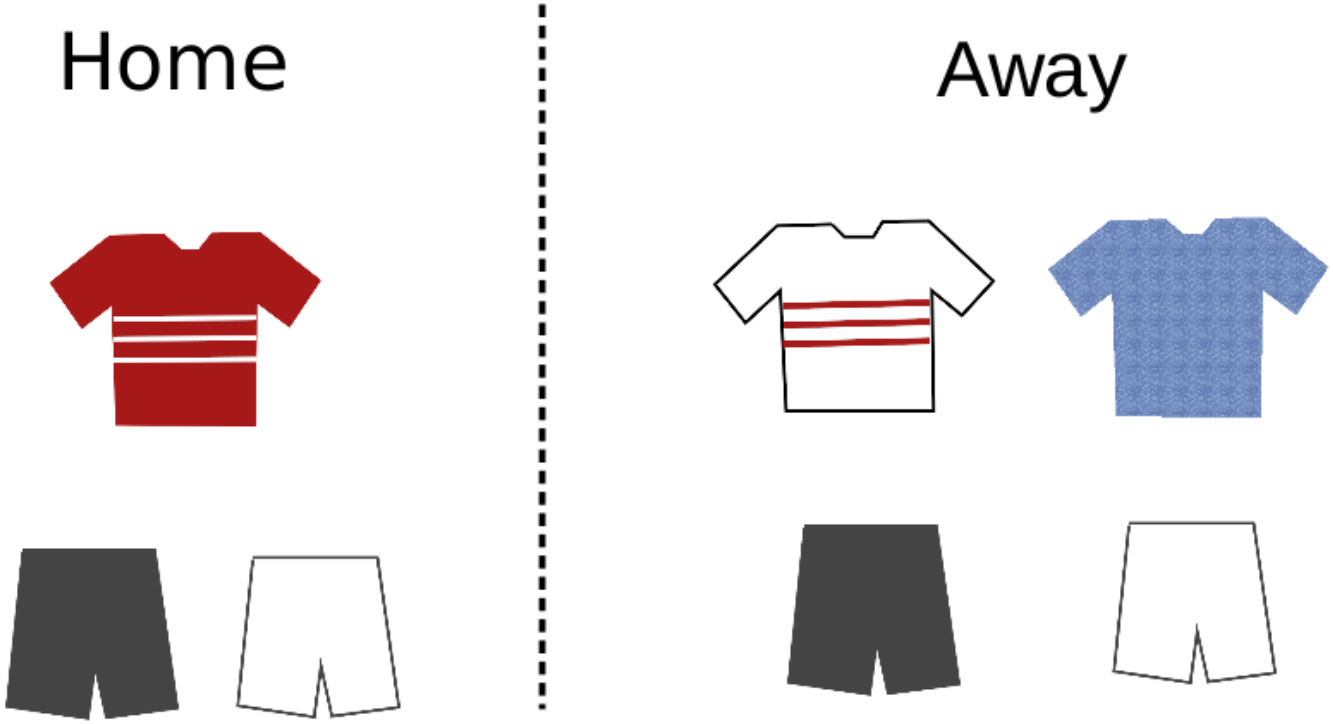
Figure: Rosen, p. 388

Each iteration increments the value of k by 1, so the final value of k is equal to the number of iterations.
Each iteration corresponds to an m-tuple $(i_1, i_2, \ldots, i_m)$ s.t. $1 \leq i_j \leq n_j$.
Ans: $n_1 n_2 \ldots n_m$.

Outline

1. General information
2. Basic principles of counting
   - The product rule
   - The sum rule
   - The subtraction rule
   - The division rule
3. Pigeon hole principle

A football club's kits



How many different kit combinations are there?

- Home: $1 \times 2 = 2$ ways;
- Away: $2 \times 2 = 4$ ways;
- Total: $2 + 4 = 6$ ways;

The sum rule

When and how does the *sum rule* apply?

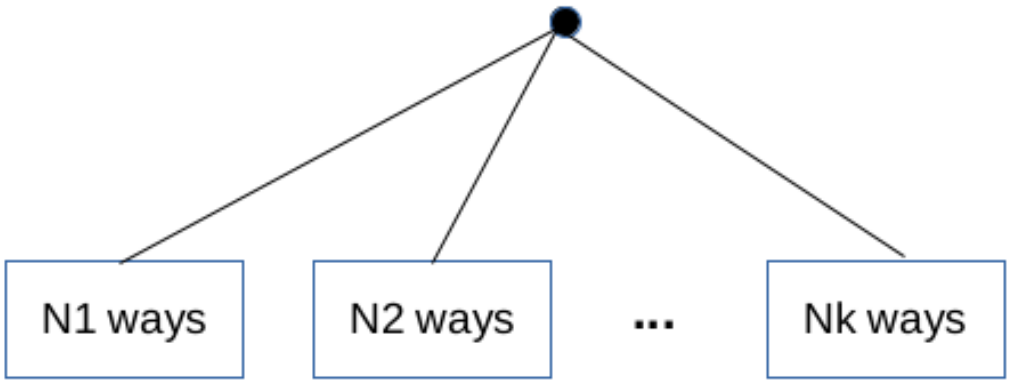
- To count the number of ways to carry out a procedure when there are *(mutually exclusive) choices*.
- The total number of ways equals the sum of the numbers of ways of each choice.
- The sum rule is usually used in combination with the product rule.

The sum rule – a formal description

Definition (Sum rule)

Suppose a procedure can be done in one of n_1 ways, or one of n_2 ways, . . . , or one of n_k ways (where the sets of n_i ways are pairwise disjoint). Then there are $n_1 + \cdots + n_k$ ways to do the procedure.

$$|A_1 \sqcup \cdots \sqcup A_k| = |A_1| + \cdots + |A_k|$$



The sum rule – examples

Example:

A forum's usernames must be between 6 and 8 character long and can only contain lowercase letters or digits. How many valid usernames are there for that forum?

Three choices:

- 6-character long usernames;
- 7-character long usernames;
- 8-character long usernames.

Let U_6, U_7, U_8 be the numbers of valid usernames of lengths 6, 7, 8, respectively. By the sum rule, the total number of valid usernames is $U_6 + U_7 + U_8$.

There are 36 choices (26 English letters and 10 digits) for each character in a username, hence by the product rule, $U_6 = 36^6$, $U_7 = 36^7$, $U_8 = 36^8$.

Ans: 2,901,650,853,888.

The sum rule – examples

Example:

A forum's usernames must be between 6 and 8 character long and can only contain lowercase letters or digits. If one valid username is chosen at random, what is the probability that it contains at least one digit?

Two choices:

- usernames with at least one digit;
- usernames without any digit.

The number of valid usernames not containing a digit is
$26^6 + 26^7 + 26^8 = 217,167,790,528$.

Hence the number of valid usernames containing at least one digit is

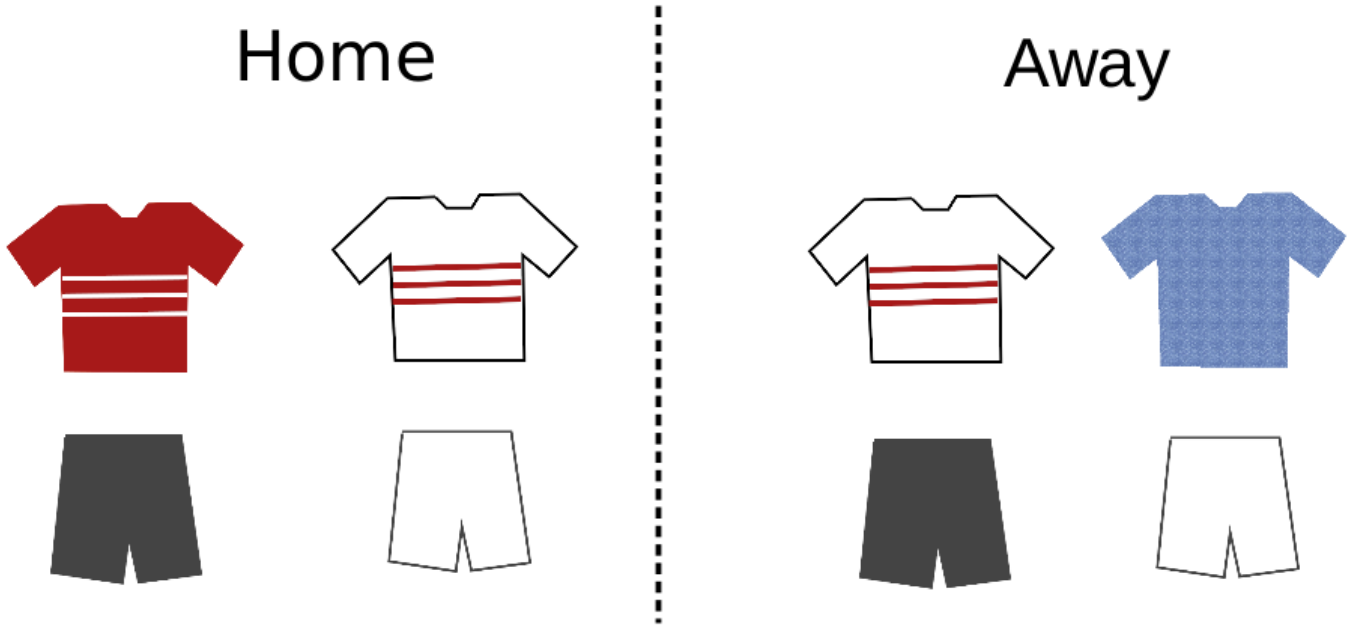
$$2,901,650,853,888 - 217,167,790,528 = 2,684,483,063,360\,.$$

Ans: 2,684,483,063,360 / 2,901,650,853,888 $\approx$ 0.925

Outline

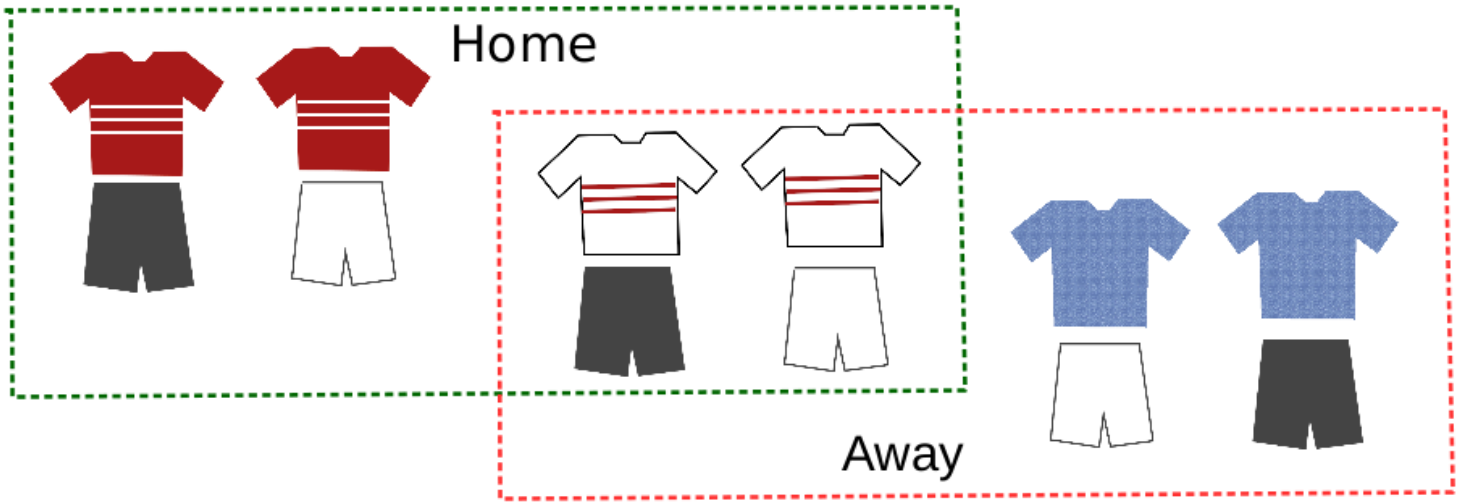
1. General information
2. Basic principles of counting
   - The product rule
   - The sum rule
   - The subtraction rule
   - The division rule
3. Pigeon hole principle

A football club's kits



How many different kit combinations are there?

A football club's kits

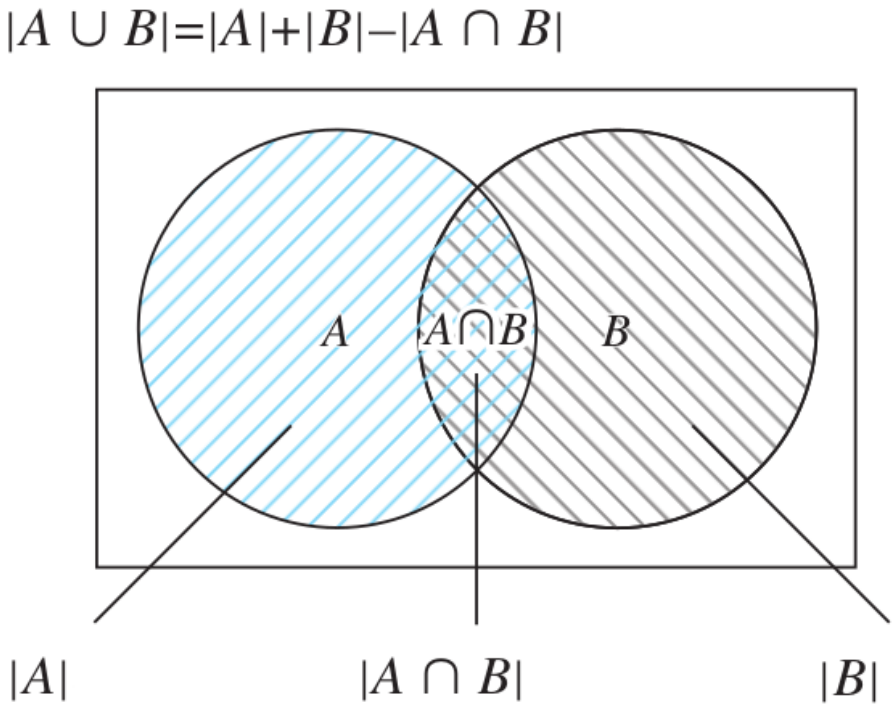


- Home: 4 ways; Away: 4 ways;
- Two common kits are counted twice!
- Total: $4 + 4 - 2 = 6$ ways.

The subtraction rule – a formal description

Definition (Subtraction rule)

$$|A \cup B| = |A| + |B| - |A \cap B|$$



The subtraction rule – example

Example:

How many binary strings of length 8 either start with a 1 bit or end with the two bits 00?

There are

- $2^7 = 128$ binary strings that start with 1;
- $2^6 = 64$ binary strings that end with 00;
- $2^5 = 32$ binary strings that start with 1 *and* end with 00.

Ans: 160

Outline

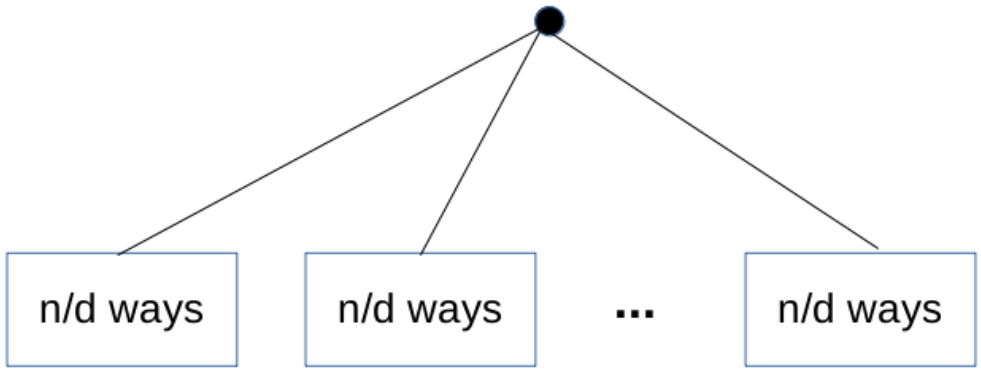
1. General information
2. Basic principles of counting
   - The product rule
   - The sum rule
   - The subtraction rule
   - The division rule
3. Pigeon hole principle

The division rule – a formal description

Definition (Division rule)

If $A = A_1 \sqcup \cdots \sqcup A_k$ *and* $|A_1| = \cdots = |A_k|$ *then*

$$|A_1| = \cdots = |A_k| = |A|/k$$



The division rule – example

Example:

If n people each shake hands with every other person, then how many handshakes are there in total?

Each person shakes hands with the others $n-1$ times, hence n people shake hands with each other $n(n-1)$ times.
But each handshake is counted twice, so the total number of handshake is $n(n-1)/2$.

The pigeon hole principle

Definition (Pigeon hole principle)

If $n+1$ pigeons stay in n pigeon holes, then there is at least one pigeon hole that accommodates more than one pigeon.

Note: The pigeon hole principle is also known as the Dirichlet's principle, the drawer principle, etc.

The pigeon hole principle – examples

Example:

- Among 13 students, there are at least two who were born in the same month.
- Among 27 arbitrary English words, there are at least two that begin with the same letter.
- A function $f : \{1, 2, \ldots, n+1\} \to \{1, 2, \ldots, n\}$ cannot be injective.

The pigeon hole principle – examples

Example:

Show that every positive integer n has a multiple that is written with only 0's and 1's.

Consider $n+1$ numbers: 1, 11, 111, ..., 11...1 ($n+1$ digits). By the pigeon hole principle, at least two among them have the same remainder when divided by n. The difference of these two number is a multiple of n that is written with only 0's and 1's.

The generalized pigeon hole principle

Definition

If n pigeons stay in k pigeon holes, then there is at least one pigeon hole that accommodates at least $\lceil n/k \rceil$ pigeons.

Note: For any real number x, $\lceil x \rceil$ is the smallest integer that is greater than or equal to x. For example, $\lceil 1.2 \rceil = 2, \lceil -1.2 \rceil = -1$.

The generalized pigeon hole principle – examples

Example:

How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

We have $k = 4$ (four suits), and we want that $\lceil n/k \rceil \geq 3$. The smallest n guaranteeing this is $n = 2k + 1 = 9$.

The generalized pigeon hole principle – examples

Example:

Assume that in a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.

